doi:10.1111/evo.13009

Diversification across a heterogeneous landscape

Greg M. Walter,[1,2] Melanie J. Wilkinson,[1] Maddie E. James,[1] Thomas J. Richards,[1] J. David Aguirre,[1,3] and Daniel Ortiz-Barrientos[1]

[1] *School of Biological Sciences, University of Queensland, St. Lucia QLD 4072, Australia*

[2] *E-mail: gregory.walter@uqconnect.edu.au*

[3] *Institute of Natural and Mathematical Sciences, Massey University, Auckland 0745, New Zealand*

Received November 25, 2015
Accepted July 5, 2016

Adaptation to contrasting environments across a heterogeneous landscape favors the formation of ecotypes by promoting ecological divergence. Patterns of fitness variation in the field can show whether natural selection drives local adaptation and ecotype formation. However, to demonstrate a link between ecological divergence and speciation, local adaptation must have consequences for reproductive isolation. Using contrasting ecotypes of an Australian wildflower, *Senecio lautus* in common garden experiments, hybridization experiments, and reciprocal transplants, we assessed how the environment shapes patterns of adaptation and the consequences of adaptive divergence for reproductive isolation. Local adaptation was strong between ecotypes, but weaker between populations of the same ecotype. F1 hybrids exhibited heterosis, but crosses involving one native parent performed better than those with two foreign parents. In a common garden experiment, F2 hybrids exhibited reduced fitness compared to parentals and F1 hybrids, suggesting that few genetic incompatibilities have accumulated between populations adapted to contrasting environments. Our results show how ecological differences across the landscape have created complex patterns of local adaptation and reproductive isolation, suggesting that divergent natural selection has played a fundamental role in the early stages of species diversification.

KEY WORDS: **Adaptation, divergence, genetic incompatibilities, heterogeneous landscape, diversification, natural selection.**

As populations colonize heterogeneous landscapes, spatial variation in natural selection and reduced migration creates strong genotype by environment interactions resulting in local adaptation (Lenormand 2002; Bridle and Vines 2007). Local adaptation to contrasting environments favors the formation of ecotypes, which may lead to broad-scale species diversification when barriers to reproduction arise and ecotypes transition to species (Lowry 2012). As genetic barriers to interbreeding are often weak, ecotypes represent a transitory stage between ecological divergence and speciation (Abbott and Comes 2007). Therefore, understanding the link between ecotype and species diversification needs to consider how reproductive isolation arises between populations locally adapting to contrasting environments. Several systems have been used to study broad-scale patterns of adaptation (e.g., Egan and Ott 2007; Hereford and Winn 2008; Gandon and Nuismer 2009; Baythavong and Stanton 2010; Baythavong 2011), but few have explored how the environment has created adaptive divergence and the consequences for reproductive isolation during diversification across a heterogeneous landscape (Angert and Schemske 2005; Vines and Schluter 2006; Sambatti and Rice 2007; Angert et al. 2008; Lowry et al. 2008b; Peterson et al. 2016). Combining reciprocal transplant experiments across multiple environments with artificial hybridization in common garden experiments can identify patterns of adaptation across a heterogeneous landscape and investigate the consequences of recent adaptive divergence, for reproductive isolation.

Reciprocal transplants that include replicate populations from multiple environments provide the opportunity to explore adaptation at different levels of biological and spatial organization (Simpson 1953; Endler 1977; Orr and Smith 1998; Schluter 2001; Fry 2003). If differences in natural selection are responsible for diversification, we expect to see stronger patterns of adaptation

© 2016 The Author(s). *Evolution* © 2016 The Society for the Study of Evolution.

between ecotypes, than between populations of the same ecotype (Hereford and Winn 2008). Across a heterogeneous landscape, populations may experience a broad range of habitat types (e.g., inland forest vs. coastal sand dunes) associated with spatial gradients of specific environmental variables (e.g., soil nutrients; Lowry et al. 2014). Generally, if populations are adapted to local conditions we expect them to perform better in transplant sites that are more similar to their own environment (Hereford 2009; Nosil 2012). Quantifying soil and climatic variables for populations included in a reciprocal transplant can test whether variation in the environment determines performance across transplant sites, providing insight into the role of natural selection in adaptive divergence across a heterogeneous landscape (MacColl 2011).

As a consequence of local adaptation, reproductive isolation can arise either dependent (extrinsic) or independent (intrinsic) of environmental differences (Coyne and Orr 2004). Reciprocal transplants that include F1 hybrids can test for extrinsic reproductive isolation such as immigrant inviability and extrinsic postzygotic isolation (Lowry et al. 2008a; Lowry et al. 2008b; Melo et al. 2014). F1 hybrids are expected to fail in the natural environment if there is selection against hybrids (Hatfield and Schluter 1999), or in any environment if intrinsic reproductive isolation reduces fitness (Coyne and Orr 2004). However , F1 hybrids often exhibit heterosis by performing better than their congenitors (Rhode and Cruzan 2005), obscuring the effect of ecological divergence on speciation.

Including multiple F1 hybrid crosses in reciprocal transplant experiments and comparing the fitness of F1 hybrids with a native parent to those with only foreign parents offers an alternative experimental design to using reciprocal backcrosses to determine the presence of extrinsic reproductive isolation in a system (Rundle and Whitlock 2001; Rundle 2002; Egan and Funk 2009; Richards et al. 2016). The native to foreign F1 hybrid contrast can account for the effects of heterosis while uncovering the ecological basis of reductions in hybrid fitness. Overall, if extrinsic postzygotic isolation has evolved between populations adapted to different environments, we expect F1 hybrids with a native chromosome to exhibit higher overall fitness than those with no native chromosome.

F1 hybrids are also expected to exhibit reduced fitness in the benign conditions of the laboratory if they have accumulated genetic incompatibilities that reduce their fitness (Seehausen et al. 2014). Nonetheless, heterosis can mask intrinsic reproductive isolation if it is weak, making it difficult to identify whether genetic incompatibilities are evolving. In the presence of heterosis, the F2 and F3 generations of hybrid populations can be used to examine the presence of genetic incompatibilities that reduce intrinsic hybrid fitness (Edmands 1999; Fenster and Galloway 2000; Fishman and Willis 2001; Johansen-Morris and Latta 2006). Genetic incompatibilities that arise following recombination can describe how intrinsic reproductive isolation is evolving during adaptive divergence across a heterogeneous landscape (Hall et al. 2010; Lowry 2012; Seehausen et al. 2014).

We investigated adaptive diversification within the recent radiation of *Senecio lautus* (Asteraceae), a wildflower species complex native to Australia, New Zealand, and several Pacific islands that occupies an array of contrasting environments across much of the Australian continent (Thompson 2005). The complex contains many taxonomic species and we focus on several varieties within *Senecio pinnatifolius* (see Roda et al. 2013a). *Senecio pinnatifolius* is an obligate outcrosser that is insect pollinated and wind dispersed. Ecotype life histories vary from annual to short-lived perennial. Previous research has shown that immigrant inviability and extrinsic postzygotic isolation between coastal ecotypes is strong and can almost fully isolate adjacent (parapatric) coastal populations (Melo et al. 2014; Richards and Ortiz-Barrientos 2016; Richards et al. 2016). Genetic differentiation between ecotypes is weak and adaptive divergence has likely evolved within the last million years (Roda et al. 2013a; Melo-Hurtado 2014). Our research used three replicate populations from four ecotypes (Ali 1964; Radford et al. 2004; Roda et al. 2013a, b; Melo et al. 2014) that occur across central Eastern Australia in coastal sand dunes (*Senecio pinnatifolius* var. *pinnatifolius*; Dune ecotype), on rocky headlands (*Senecio pinnatifolius* var. *maritimus*; Headland ecotype), moist tableland rainforest (*Senecio pinnatifolius* var. *serratus*; Tableland ecotype) and in dry woodland (*Senecio pinnatifolius* var. *dissectifolius*; Woodland ecotype). The two coastal ecotypes are parapatric along the coastline, and the two inland ecotypes occupy habitats in close spatial proximity of each other, so dispersal between habitats occupied by divergent ecotypes is possible. To study patterns of local adaptation within and between ecotypes we conducted a reciprocal transplant of all 12 populations and F1 hybrids into the four environments. We also quantified multiple soil and climatic variables for each population to identify aspects of the environment that may predict patterns of adaptation in the system. We then used common garden experiments to create advanced hybrid generations and quantify intrinsic fitness to assess whether intrinsic reproductive isolation has evolved in the system.

Materials and Methods

SAMPLE COLLECTION

We sampled seeds from 12 natural populations from four ecotypes of *S. pinnatifolius* (Table S1). Seeds were collected from 30 individuals that were a minimum of 10 meters apart to reduce the risk of collecting closely related individuals. To quantify the environment for each population we took soil samples, which we submitted to the Australian Laboratory Services to analyze the soil composition for metals (29 variables), nutrients

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(11 variables including nitrogen and carbon content), salt (chloride), pH, cation exchange capacity (and levels of exchangeable cations), and electrical conductivity. Soil data from the coastal populations were previously used for genetic analyses in Roda et al. (2013b). For each population we extracted six temperature and precipitation climate variables from the WordClim dataset (http://www.worldclim.org/): mean diurnal temperature range, mean temperature of warmest quarter, mean temperature of coldest quarter, precipitation of driest month, precipitation of warmest quarter, and precipitation of coldest quarter. Digitech weather stations were placed at each transplant location to capture variation in microclimate including humidity, wind, and temperature between the transplant sites. To characterize the difference between ecotype environments, we used multidimensional scaling on the climate and soil variables to calculate the distance between populations in environmental space (Anderson 2006). To test whether ecotype environmental centroids were significantly different, we used a permutation test from the R package "vegan" (Oksanen et al. 2016).

CREATING CROSS TYPES AND SEED RESOURCES

Seeds from each of the 12 populations were germinated and grown in University of Queensland glasshouses for one generation to reduce nonglasshouse maternal effects and produce parental and hybrid seeds for the transplant experiment (Bischoff and Muller-Scharer 2010). Germination was induced in the laboratory by scarifying seeds and then placing them in petri dishes lined with moist filter paper. Petri dishes were kept in the dark for 2 days before being transferred to a controlled temperature room where light was maintained at a 50:50 day-night cycle at 25°C. After one week, plants were transferred to the glasshouse where they were transplanted into 85 mm square pots containing soil (70% pine bark: 30% coco peat) with 5 kg/m^3 slow release osmocote fertilizer and 830 g/m^3 Suscon Maxi insecticide. When plants commenced flowering, controlled crosses were conducted by rubbing two flower heads together repeatedly over several days, allowing all flowers within each inflorescence to donate and accept pollen. Crossing tags were used to track fertilized flower heads and seeds were collected in seed envelopes once they emerged.

To create the hybrid generations we grew plants from one population for each ecotype and crossed them to produce all 12 combinations of F1 hybrids. The number of families used for the Dune, Headland, Tableland, and Woodland populations were 51, 53, 35, and 24, respectively. Although a different number of families were used for each population, only 25 families for each F1 cross type (HeadxDune, DunexHead, HeadxTable, TablexHead, and so forth) were used to produce the F2 generation. F1 hybrids were crossed to produce the F2 generation in such a way that each cross contained 25% of the genome from each ecotype (i.e., HxD was crossed with WxT). Finally, F3's were created by randomly crossing the F2's while making sure that each F2 cross type was equally represented. A separate population of F2 hybrids was also created in a similar way between one Dune and one Headland population (both from Lennox Head, NSW) to determine the effect of recombination between four ecotypes versus two ecotypes. Table S2 contains the number of full-sibling families and individuals used to create the two types of advanced hybrid generations.

RECIPROCAL TRANSPLANT EXPERIMENT

We conducted the reciprocal transplant experiment in four contrasting environments in southern Queensland and northern New South Wales (NSW). Transplant sites included a coastal sand dune (dune environment; S 28° 47′ 1.23″, E 153° 35′ 38.56″) and coastal rocky headland (headland environment; S 28° 48′ 47.22″, E 153° 36′ 19.15″) at Lennox Head (NSW), inland sclerophyll woodland (woodland environment; S 28° 20′ 19.82″, E152° 2′ 23.75″) near Warwick (Queensland), and warm-temperate rainforest on a high plateau (tableland environment; S 28° 21′ 15.55″, E 152° 23′ 48.44″) near Killarney (Queensland). The only transplant site that did not include the local population was the woodland site, which was located only after all Woodland seeds were produced in the glasshouse. All seeds were transplanted into the four environments on the 18th and 19th of March 2014 (i.e., the start of the Australian autumn).

At each transplant site 12 parental populations and six F1 crosses were randomized into six blocks of 500 seeds each, totalling 12,000 seeds for the transplant. To prepare the seeds we used nondrip superglue to glue each seed to a toothpick and stored them in boxes in a preprepared randomized grid that replicated the grid setup in the field. In the field, toothpicks with seeds were planted individually into 25 mm square grid cells so the seed sat 1–2 mm below the soil surface (Fig. S3). Shade cloth (50%) was suspended 15 cm above the plots. Plots were kept moist for the first 3 weeks to induce germination by replicating natural germination conditions. During this three-week period emergence and mortality was recorded every second day. Following this, data were collected in weeks 4, 5, 7, and 9, and then monthly for 11 months until mortality in all environments dropped below 20%. We recorded emergence, mortality, whether plants reached seedling establishment (produced 10 leaves) and whether plants reached maturity (produced a bud). However, very few plants produced a bud, and insufficient data meant the ability for plants to reach maturity was omitted from analyses. All measurements were recorded as binary variables. Shade cloth was replaced with very light bird netting once germination ceased, which was after 30 days for all environments except the Dune plots, which was replaced after 100 days due to the harshness of the environment and to replicate native *growing* conditions for Dune plants under spinifex grass, rather than open sand. Bird netting provided a

small amount of shade and was used to ameliorate dramatic water loss that would be associated with clearing of the ground cover prior to sowing seeds. The bird netting was completely removed after 100 days in the Headland, Tableland, and Woodland environments but was retained in the Dune environment until the end of the experiment to replicate growing under spinifex grass.

DATA ANALYSIS

Initially, we estimated differences in survival for each ecotype in each environment using the R package "coxme" (Therneau 2015). Time to death was analyzed within each environment where ecotype was fitted as a fixed effect, with replicate population and environmental block included as random effects. We reordered the ecotypes so that the native ecotype became the intercept in the model for each environment, allowing comparison of the native ecotype with all foreign ecotypes.

For the analyses of the reciprocal transplant data we used a bivariate generalized linear-mixed model in a Bayesian framework using the R package "MCMCglmm" (Hadfield 2010). The R code used to produce the models is located in Supporting Material S4. To estimate the performance of each ecotype in each environment we used the following model

$$\mathbf{y}_{\mathrm{ijklmn}} = \mathbf{E}_{\mathrm{i}} + \mathbf{O}_{\mathrm{j}} + \mathbf{EO}_{\mathrm{ij}} + \mathbf{P}_{\mathrm{k(j)}} + \mathbf{F}_{\mathrm{l(jk)}} + \mathbf{B}_{\mathrm{m(i)}} + \mathbf{e}_{\mathrm{nijklm}}, \quad (1)$$

where transplant environment ($\mathbf{E}_{\mathrm{i}}$), the ecotype origin of seeds ($\mathbf{O}_{\mathrm{j}}$), and interaction ($\mathbf{EO}_{\mathrm{ij}}$) were fitted as fixed effects. Population within ecotype ($\mathbf{P}_{\mathrm{k(j)}}$), family within population ($\mathbf{F}_{\mathrm{l(jk)}}$), and environmental block ($\mathbf{B}_{\mathrm{m(i)}}$) within transplant environment were fitted as random effects and $\mathbf{e}_{\mathrm{nijklm}}$ was the residual error. To compare F1 and parental ecotype performance we included parental and F1 hybrids in the $\mathbf{O}_{\mathrm{j}}$ term of a second model.

To investigate patterns of local adaptation using population performance (regardless of ecotype) across environments we used

$$\mathbf{y}_{\mathrm{ijklm}} = \mathbf{E}_{\mathrm{i}} + \mathbf{P}_{\mathrm{j}} + \mathbf{EP}_{\mathrm{ij}} + \mathbf{F}_{\mathrm{k(j)}} + \mathbf{B}_{\mathrm{l(i)}} + \mathbf{e}_{\mathrm{mij}}, \quad (2)$$

where transplant environment ($\mathbf{E}_{\mathrm{i}}$), population ($\mathbf{P}_{\mathrm{j}}$), and their interaction ($\mathbf{EP}_{\mathrm{ij}}$) were fitted as fixed effects. Family within population ($\mathbf{F}_{\mathrm{k(j)}}$) and environmental block ($\mathbf{B}_{\mathrm{l(i)}}$) within transplant environment were fitted as random effects and $\mathbf{e}_{\mathrm{mij}}$ was the residual error. The bivariate model was biologically meaningful for our data as it estimated performance for a fitness component conditional on previous life-history traits (Steinsland et al. 2014). As many plants failed to reach maturity, we used the latest life-history stage with sufficient data to represent our best estimate of overall fitness, which was seedling establishment (ability to reach 10 leaves). Seedling establishment was modeled conditional on germination and survival to day 100. Survival to day 100 was included as the survival component preceding seedling establishment and was estimated as the presence or absence of a plant when data were collected at 100 days. The three field measurements of performance were entered as a vector of response variables in model 1 ($\mathbf{y}_{\mathrm{ijklmn}}$) and model 2 ($\mathbf{y}_{\mathrm{ijklm}}$), with a dummy variable identifying the life-history traits included as a fixed effect.

All models were run for 1,100,000 Markov chain Monte Carlo (MCMC) sampling iterations with a burn-in of 100,000 iterations and a thinning interval of 1,000 iterations. A Cauchy prior distribution was used for all random effects (Gelman 2006). To test the influence of the prior on our estimates, we repeated the analyses while adjusting the scale parameter of the Cauchy distribution to excessively large and excessively small values, making sure it did not change the model output. The number of iterations and the thinning interval were increased until the effective sample size exceeded 85% of the total number of samples for all parameters estimated. We checked for autocorrelation between MCMC samples to validate the performance of the chain. Model performance was verified using posterior predictive checks comparing the posterior distribution against the distribution of the raw data. All models indicated a precise alignment between the two distributions. Significance testing in a Bayesian framework is conducted by assessing whether the MCMC distributions for the two contrasts under comparison overlap. All figures presented below include the 95% highest posterior density (HPD) intervals. No overlap between two distributions indicates evidence for a significant difference between the distributions.

LOCAL ADAPTATION CONTRASTS

We performed traditional tests of local adaptation for the four ecotypes across the four environments. We considered ecotypes locally adapted when they performed better in their own environment than foreign ecotypes ("Local vs. Foreign" or LF criteria) and better in their own environment than in alternative environments ("Home vs. Away" or HA criteria; Kawecki and Ebert 2004). Then, to understand broad-scale patterns of adaptation across a heterogeneous landscape we investigated adaptation both within and between ecotypes. Specifically, we compared "Local" populations planted in their local environment (e.g., Dune population from Lennox Head planted in the dune at Lennox Head), "Native" but nonlocal populations (e.g., nonlocal Dune populations planted in the Lennox Head dune environment) and "Foreign" populations, which refers to nonlocal populations in a nonnative environment (e.g., Tableland populations planted in the Lennox Head dune environment). We refer to this contrast as the "Local-Native-Foreign" (LNF) contrast.

ROLE OF THE ENVIRONMENT

We calculated environmental distance as the distance between each population and the transplant locations using Euclidean distance in environmental space from the multidimensional scaling

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

analysis outlined previously. Linear regressions of environmental distance against population performance in each transplant environment (eq. 2) were then used to examine the relationship between environment distance from the transplant site, and field performance. Following this, we used the measured environmental variables to investigate whether specific environmental variables were associated with field fitness. The first principal components for soil fertility, macronutrients, micronutrients, and metals were highly correlated with each other, shown by Pearson's correlation coefficients greater than 0.9 for all comparisons (Fig. S5). Consequently, we used the principal component analysis for soil fertility to represent soil composition. The first principal component for soil fertility and climate described 59.9% and 60.1% of total multivariate soil and climate variance, respectively. We used the first principal components as the new variables representing soil composition and climate. Next, we regressed the first principal components for soil fertility and climate, and the raw data for soil salt (chloride) and soil pH environmental variables onto field performance in each environment, allowing us to test for associations between environmental variables and field performance. Figure S6 contains the regressions for the second principal components of soil fertility and climate. To calculate uncertainty around each regression model, we randomly sampled the distribution of the explanatory variable and regressed the random samples against an MCMC iteration of population performance. We then repeated the randomization and regression for 500 MCMC iterations. The slope for each regression was determined to be significant if the distribution of 500 estimates of the slope did not overlap zero.

ESTIMATING EXTRINSIC REPRODUCTIVE ISOLATION

Using the reciprocal transplant data, we calculated the strength of immigrant inviability as $RI_{imm} = 1 - (w_i/w_n)$, where w_i was the mean survival of foreign ecotypes, and w_n was the native ecotype. We also calculated hybrid inviability as $H_{hi} = 1 - (v_{mean}F1/v_{native})$, where v_{mean} was the survival of the hybrids with a native parent, and v_{native} was the native ecotype (Lowry et al. 2008a; Melo et al. 2014). Finally, we calculated the strength of selection against hybrids with foreign parents using $H_{hc} = 1 - (v_{foreign}F1 / v_{native}F1)$, where $v_{foreign}F1$ was the survival of the hybrids with foreign parents, and $v_{native}F1$ was the hybrids with a native parent.

ESTIMATING INTRINSIC REPRODUCTIVE ISOLATION

We estimated intrinsic fitness of hybrid genotypes by measuring fertilization success of crosses conducted in the glasshouse. Crosses were considered successful when at least one viable seed was produced, which provided binary data for crossing success (0 = failed cross, 1 = successful cross). In this system, viable seeds are round and range from black to light gray, while nonviable seeds are easily distinguished as white, thin, and hollow. For the crosses that produced seeds we counted the number of viable and nonviable seeds and expressed these as proportions (viable seeds/total seeds) to provide a measure of fertilization success. We measured intrinsic fitness in parentals and F1, F2, and F3 hybrid generations.

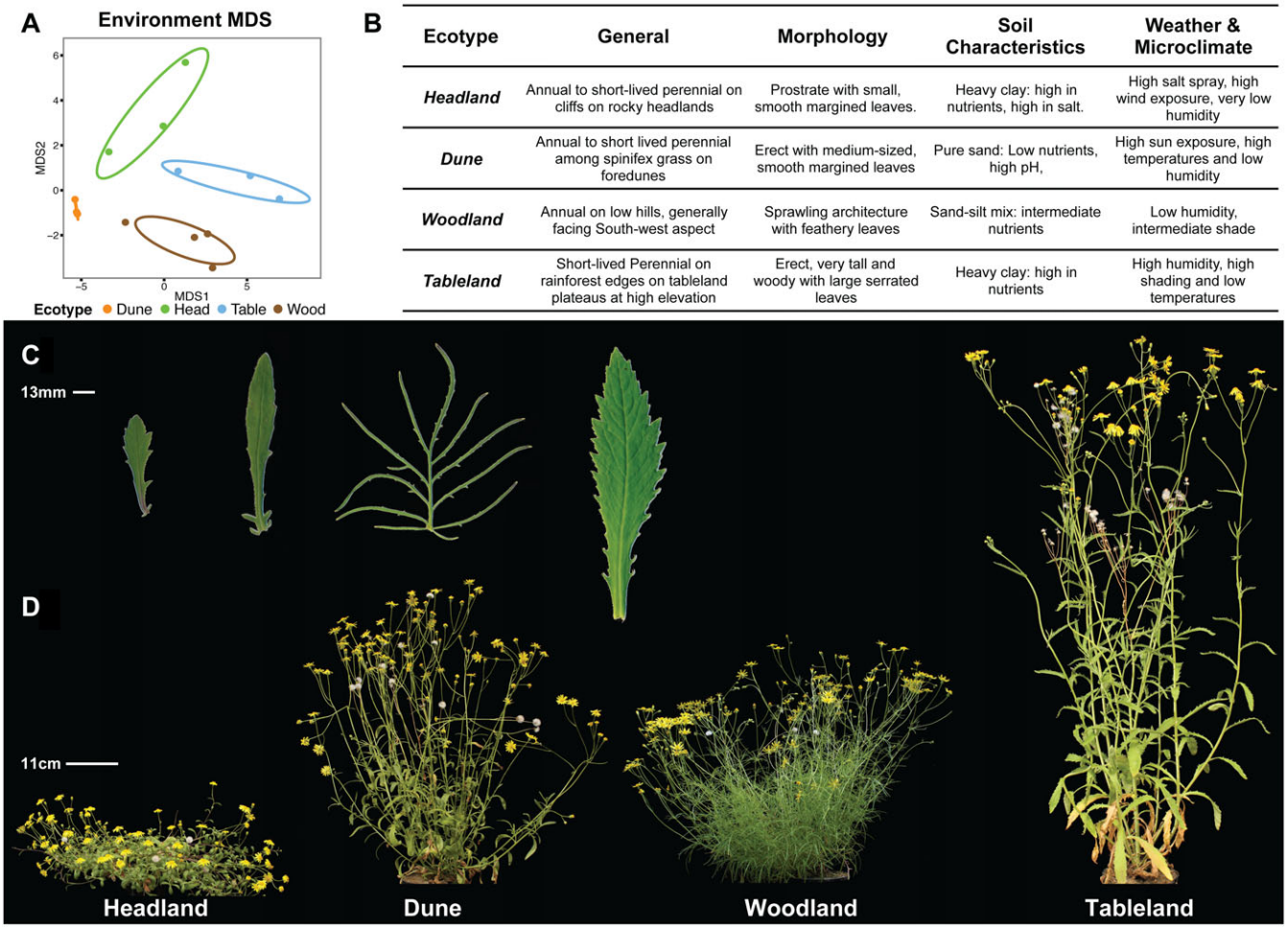
Results

CHARACTERIZING THE ECOTYPES AND THEIR ENVIRONMENTS

The multivariate analysis of soil and climate environmental variables revealed clear differences between environments, with populations from each ecotype occupying similar environments that group together and well away from locations representing other ecotypes (Fig. 1A). Centroids representing ecotype environments were significantly different from each other (permutation test; $n = 999$, $R^2 = 0.772$, $P = 0.001$). Ecotypes differed in growth habit, morphology, and occupied environments that differ greatly in micro and macro-environment (Fig. 1B). Figure S7 illustrates the microenvironment differences captured by the weather stations. Overall, the Headland site had very low humidity and dewpoint but very high wind speeds. The Dune site was characterized by higher temperatures and a higher dewpoint. Tableland had the highest humidity and low temperatures while the Woodland site had the lowest temperatures and intermediate humidity (between the Tableland and Headland environments). Differences in environment were reflected by contrasting phenotypic differences in leaf morphology (Fig. 1C) and plant architecture (Fig. 1D).

RECIPROCAL TRANSPLANT EXPERIMENT

Seedling emergence across all cross types was high for all environments (Dune 70%, Headland 71%, Tableland 78%, Woodland 76%). Results of the survival analysis showed that native ecotypes survived better than foreign ecotypes in all environments, except the woodland site (Fig. 2). Furthermore, all foreign ecotypes crashed at a similar time-point in each environment (Fig. 2). Signatures of local adaptation were evident using the LF and HA criteria. Overall, we found that the Dune, Headland, and Tableland ecotypes performed significantly better in their native environment than foreign ecotypes (LF; Fig. 3A) and also performed significantly better in their native environment than alternative environments (HA; Fig. 3B). These results suggest strong local adaptation in the *S. pinnatifolius* system with the exception of the Woodland ecotype, which exhibited weak evidence of local adaptation as it performed marginally better in its local environment compared to foreign populations (Fig. 3A), and equally poorly across habitats (Fig. 3B).



Ecotype	General	Morphology	Soil Characteristics	Weather & Microclimate
Headland	Annual to short-lived perennial on cliffs on rocky headlands	Prostrate with small, smooth margined leaves.	Heavy clay: high in nutrients, high in salt.	High salt spray, high wind exposure, very low humidity
Dune	Annual to short lived perennial among spinifex grass on foredunes	Erect with medium-sized, smooth margined leaves	Pure sand: Low nutrients, high pH,	High sun exposure, high temperatures and low humidity
Woodland	Annual on low hills, generally facing South-west aspect	Sprawling architecture with feathery leaves	Sand-silt mix: intermediate nutrients	Low humidity, intermediate shade
Tableland	Short-lived Perennial on rainforest edges on tableland plateaus at high elevation	Erect, very tall and woody with large serrated leaves	Heavy clay: high in nutrients	High humidity, high shading and low temperatures

Figure 1. Environment and morphological characterization of the four ecotypes. (A) Multidimensional scaling analysis of soil and climate variables shows a strong separation between ecotypes and grouping of populations within an ecotype. (B) General description of the environments, growth habit, morphology, and microenvironment. This table was constructed from the environmental data, the weather station data, and personal observations over the transplant. The four ecotypes show strong differences in both leaf morphology (C) and plant architecture (D).

"LOCAL-NATIVE-FOREIGN" (LNF) CONTRAST OF LOCAL ADAPTATION

Using the LNF contrast (Fig. 4A), we found a strong signal of local adaptation across the system where local populations performed significantly better than foreign populations (Fig. 4B). Furthermore, nonlocal populations from the native ecotype exhibited intermediate performance between local and foreign populations, suggesting some level of local adaptation within ecotypes. However, there was considerable variance in performance of nonlocal but native populations (visualized by the colored dots in Fig. 4B). To investigate local adaptation within ecotypes we estimated field performance for local populations versus nonlocal populations from the same ecotype, in their native environment (Fig. 4C). Although there was an overall signature of local adaptation this was largely driven by the Headland ecotype, which exhibited a significantly higher local advantage over nonlocal Headland populations. Tableland also indicated a similar trend but there was no significant difference between local and native population performance.

DIFFERENCES IN THE ENVIRONMENT CREATE PATTERNS OF LOCAL ADAPTATION

We used the climate and soil data to assess how environmental distance between populations and transplant sites affected fitness. If differences in the strength of divergent natural selection created patterns of local adaptation we expected the further away a population was derived in environmental space, the more poorly it would perform in any given transplant environment (Hereford 2009). Consistent with this prediction, Figure 5 shows a strong negative correlation between environmental distance and field fitness, R^2 ranged between 0.29 and 0.67. Slopes for the regression in each environment were significantly negative, as

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

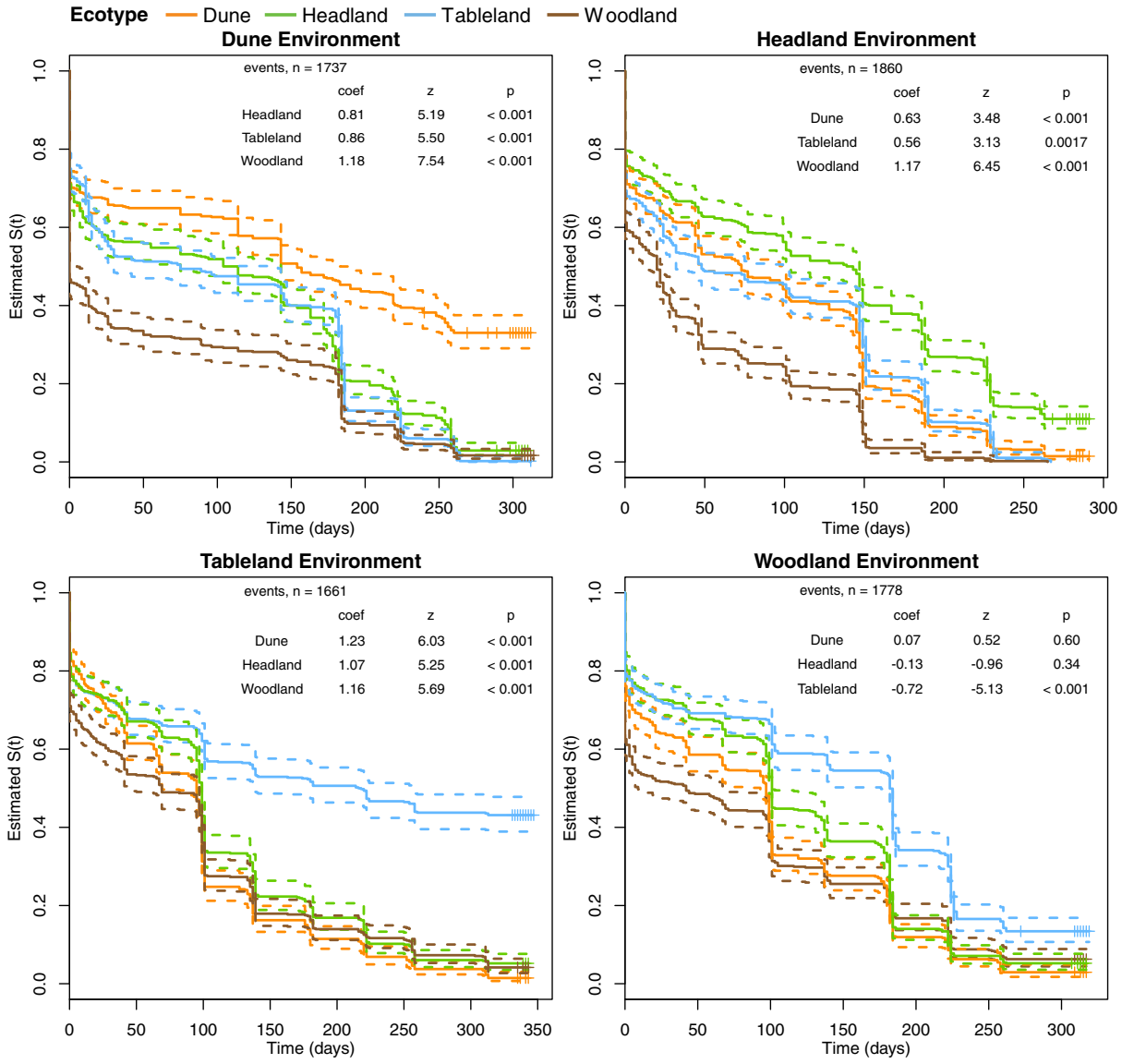


Figure 2. **Survival analysis for all ecotypes planted in each environment, with replicate populations included as a random effect. Dashed lines are 95% confidence intervals for each ecotype. The native ecotype survived much better than foreign ecotypes in all environments, except for the woodland environment where the Tableland ecotype performed better. Summary tables for Cox proportional hazard models in upper right of each panel shows differences from the native ecotype were significant.**

the posterior distribution for all slope estimates did not overlap zero. Importantly, geographic distance explained less variation in fitness across environments with very low R^2 values, ranging between 0.01 and 0.17 (Fig. S8).

ENVIRONMENTAL VARIABLES UNDERLYING ADAPTIVE DIVERGENCE

To investigate the specific environmental variables that explained variation in field fitness we tested the relationship between environmental variables and field performance (Fig. 6). Generally, we expected climate and soil variables to associate similarly with field fitness for the two coastal and two inland environments. Field performance in most environments was associated with environmental variables, illustrated by large R^2 values (Fig. 6) and regression slopes significantly different to zero (Fig. S9). The only regression slopes that were nonsignificant were soil fertility in the headland environment and salt in the two inland environments. Climate and pH exhibited an opposite response for the coastal versus inland environments (Fig. 6). Soil fertility had a strong relationship to field fitness in each environment where populations with higher soil fertility were associated with higher fitness in the two inland environments, but a reduction in the sand dune environment (Fig. 6). In contrast, increases in pH (more alkaline soils) resulted in increased performance in the dune and headland

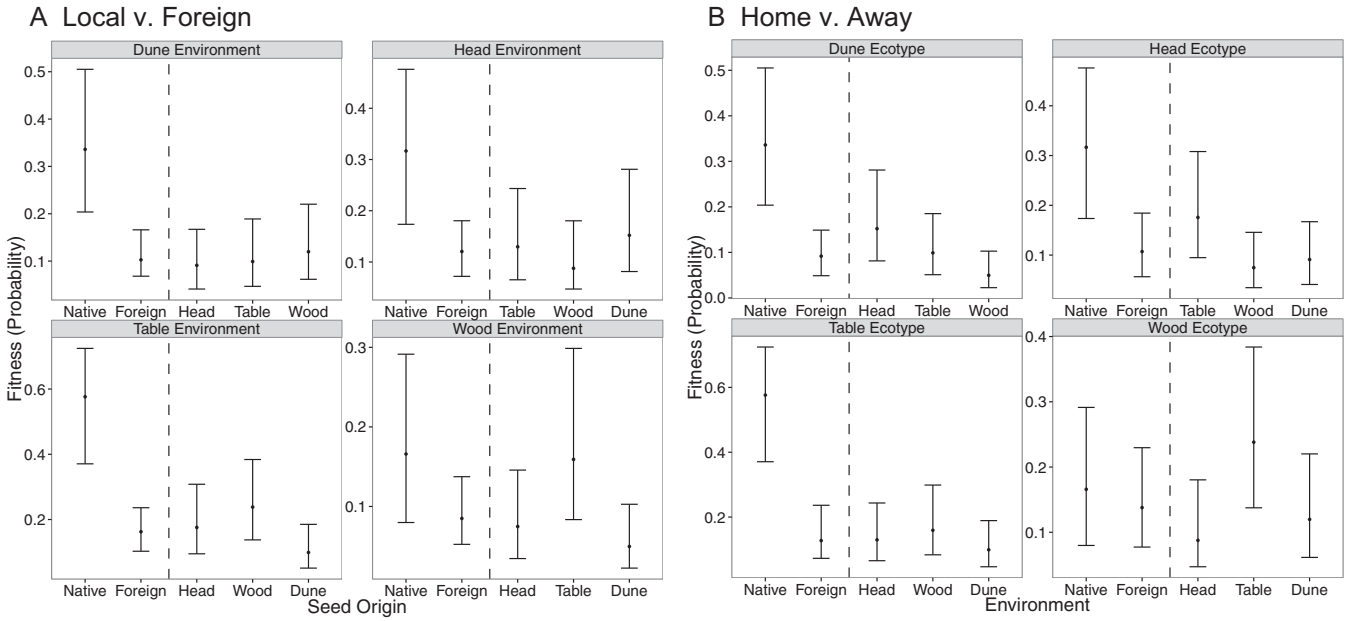


Figure 3. **(A) The "Local versus Foreign" comparison of local adaptation between ecotypes within each environment. Native is the native ecotype and foreign is an average of all foreign ecotypes. Native ecotypes performed better than foreign ecotypes. (B) The "Home versus Away" comparison where the performance of an ecotype is gauged across all environments. Native refers to an ecotype in its own environment, while foreign indicates a summary of that ecotype across the other environments. Ecotypes performed better in their own environment than foreign environments. Confidence intervals are 95% HPD intervals.**

environments, but reduced fitness in the tableland environment. Soil salt content did not predict fitness in the inland environments, however higher salt was associated with higher fitness in the headland, but lower fitness in the dune, suggesting a salt related trade-off between the two coastal sites (Fig. 6).

CONSEQUENCES OF ADAPTIVE DIVERGENCE: EXTRINSIC REPRODUCTIVE ISOLATION

Immigrant inviability was very strong for all environments, except in the woodland (RI_{imm}; dune = 0.95, headland = 0.95, tableland = 0.91, woodland = 0.08). We tested whether F1 hybrids with native parents performed better than F1 hybrids with only foreign parents (Fig. 7A). All F1 hybrids exhibited heterosis whereby they performed significantly better than their midparents (Fig. 7B). Although estimates of extrinsic hybrid inviability were weak when considering F1 performance against the native ecotype (H_{hi}; dune = 0.07, headland = 0.22, tableland = 0.23, woodland = –0.42), there was strong selection against F1 hybrids with foreign parents (Fig. 7A), except in the woodland environment (H_{hc}; dune = 0.69, headland = 0.89, tableland = 0.69, woodland = -0.30) (Fig. 7B).

CONSEQUENCES OF ADAPTIVE DIVERGENCE: INTRINSIC REPRODUCTIVE ISOLATION

We assessed intrinsic fitness of later generation experimental hybrids by measuring crossing success in the common garden experiment. The percentage of successful crosses and the proportion of viable seeds produced per cross was very high for crosses performed within the parental, F1 hybrid and F3 hybrid generations. For each generation over 90% of crosses were successful with the proportion of viable seeds ranging between 60% and 80% (parentals: 77 ± 1% (n = 82), F1 hybrid: 61 ± 2% (n = 65) and F3 hybrid: 72 ± 1% (n = 128)). However, we observed a drastic reduction in F2 fitness with a crossing success of 42% and seed set of 12 ± 1% (n = 163). Similarly, F2 hybrids produced between only the Dune and Headland populations exhibited a similar reduction in flowering fitness where only 24.6% of plants flowered (n = 744) and the individuals that managed to flower only produced 2.88 flowers on average (SD ± 3.17 flowers). In contrast, parentals and F1 hybrids all flowered and produced a minimum of 15 flowers per individual plant (G. M. Walter, pers. obs.).

Discussion

Here, we provide strong evidence for local adaptation both within and between contrasting ecotypes of *S. lautus* driving diversification across a heterogeneous landscape. Differences in field performance among populations depended on environmental differences between natural habitats, suggesting that divergent natural selection has created ecotypic divergence. More specifically, our results show that local adaptation has created almost

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

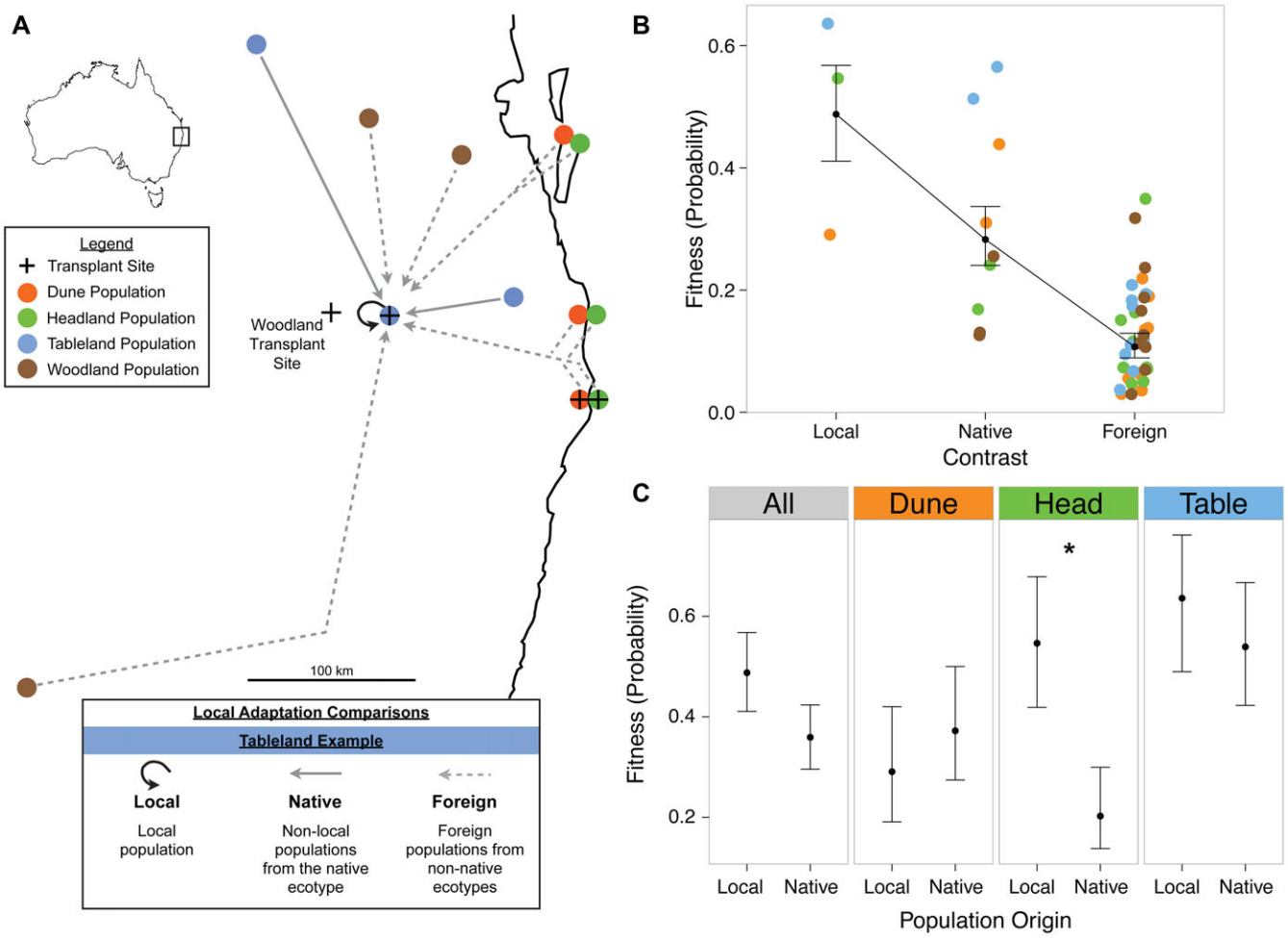


Figure 4. **(A) Test of local adaptation using the LNF contrast across the system. Local refers to the performance of local populations in their local environment, native refers to nonlocal, but native populations (i.e., native ecotype, but from a different locality) and foreign refers to populations from a foreign ecotype. The map illustrates the comparison for just the tableland environment as an example, however the full analysis includes all four environments. (B) Comparison of local populations with native (but nonlocal populations) and foreign populations across all four environments. Local populations performed better than native but nonlocal populations, which performed better than populations from foreign ecotypes, providing strong evidence for local adaptation both between and within ecotypes. Colored dots illustrate the fitness of each population and although there is a large amount of variation this is inherent to analyses of performance between contrasting environments. However, the important comparison is between the same colored dots, which show the trends within an ecotype. (C) To understand local adaptation within ecotypes we compared performance of local populations with nonlocal populations from the same ecotype, in their native environment. The analysis included the three environments that included local populations, with the left-hand column summarizing the overall trend. Local adaptation was present between populations within ecotypes overall, but the level of local adaptation depended on the ecotype. The Headland populations exhibited strong local adaptation, whereas Tableland populations exhibited only a weak signal of local adaptation and Dune populations no signal. All confidence intervals are 95% HPD intervals and the asterisk represents no significant overlap between distributions.**

complete immigrant inviability between environments, except for the Woodland ecotype, suggesting that natural selection has played a major role in the evolution of *S. lautus*. Consistent with this observation, although F1 hybrids exhibited heterosis, natural selection favored hybrids with native genes under field conditions. In the common garden experiment, F1 and F3 hybrids performed well with fitness reductions only observed in the F2 generation, suggesting that few genetic incompatibilities have accumulated in the system. Overall, our results suggest that the colonization, and subsequent adaptation to contrasting environments has created diversification across a heterogeneous landscape, providing insight into the ecological processes underlying adaptive diversification.

LOCAL ADAPTATION

Adaptive genetic variation has been identified using reciprocal transplants across local (e.g., Antonovics and Bradshaw 1970)

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

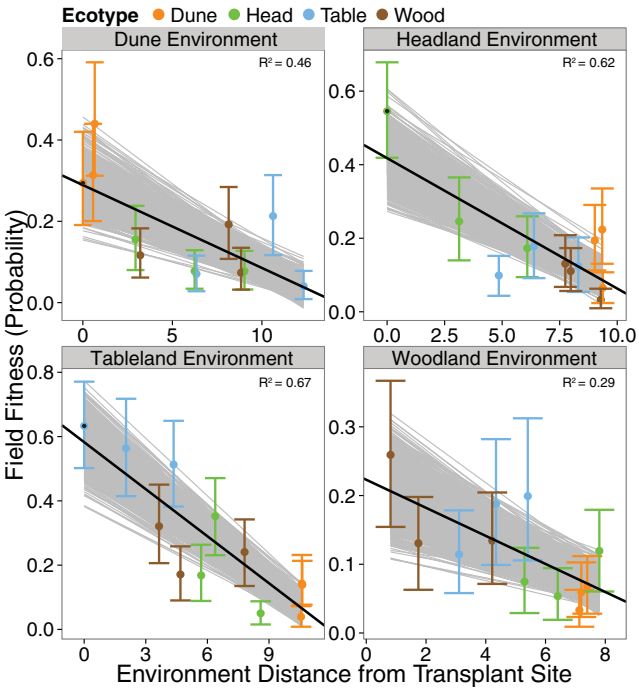


Figure 5. Environmental distance was negatively correlated with population performance. Environmental distance calculated as the Euclidean distance between each population and the transplant site in multivariate environmental space, black dots denote the local population. The solid lines indicate the point estimate of the slope and intercept in each environment. Each gray line is a separate regression between a random range of the explanatory variable and the regression model for each MCMC iteration, providing the uncertainty around the point estimates. All slopes are significantly different from zero.

and broad geographic scales (e.g., Joshi et al. 2001; Macel et al. 2007; Ågren and Schemske 2012). Since the seminal work of Clausen, Keck, and Hiesey (reviewed in Lowry 2012) several studies have explored adaptive divergence in taxa that exhibit both local adaptation and broad-scale ecotypic divergence. For instance, Hereford and Winn (2008) investigated the limits to local adaptation for three widespread habitat types of *Diodia teres*, including two replicate populations for each habitat in a full reciprocal transplant. Although plants were adapted to their own habitat, no local adaptation was evident between populations within a habitat type, suggesting that local adaptation between populations within an ecotype has only occurred where populations have adapted to locations distinctly different in environment, despite being from the same habitat type.

Our results are consistent with Hereford and Winn (2008), where ecotypes from patchily distributed yet contrasting environments exhibited strong patterns of adaptation (Figs. 3 and 4). However, local adaptation between populations within ecotypes was only partial: local populations of the Headland (and to some extent, Tableland) ecotype performed better than nonlocal populations. The small patch size of suitable habitat combined with differences in selection between rocky headlands and the difficulty of migration between them has likely contributed to these patterns of local adaptation. In contrast, the remaining three ecotypes occupy habitats where suitable environment is less spatially fragmented, possibly allowing higher levels of connectivity and reducing local adaptation between populations. The Woodland ecotype performed poorly in their native habitat, but the woodland environment is ephemeral and by inducing germination we likely exposed the woodland plants to environmental conditions they would not have naturally experienced.

ENVIRONMENTAL VARIABLES UNDERLYING PATTERNS OF ADAPTATION

Local adaptation has been described in numerous systems, but identifying agents of selection underlying patterns of adaptation is difficult. In *Mimulus*, salt creates strong selection against immigrants to coastal sites (Lowry et al. 2009), while seasonal water availability selects against immigrants to inland sites (Lowry et al. 2008b). Furthermore, edaphic conditions drive selection against immigrant *Mimulus* populations inhabiting serpentine and nonserpentine soils (Selby 2014). Our transplant of populations derived from a heterogeneous landscape showed that the edaphic and climatic environment of origin determined population performance in any given transplant environment, suggesting that soil and climatic variables were potentially responsible for differences in natural selection underlying patterns of adaptation.

If populations are adapted to their local environment, population performance across a landscape should be determined by the similarity between its native environment, and all other environments (Hereford 2009). Consistent with this prediction we noted two major patterns. First, populations adapted to high levels of an environmental variable displayed reduced fitness in environments with lower levels of the same variable. For instance, soil fertility was highly correlated with fitness in each environment: Tableland populations derived from high nutrient soils performed poorly in low nutrient soils (dune environment) and had an intermediate performance in environments with intermediate levels of soil fertility; the reverse pattern was evident for Dune populations derived from low nutrient soils, which performed poorly in high nutrient soils (Fig. 6). Second, we found a strong negative correlation between environmental distance and population fitness, which was particularly pronounced in the Tableland site. Overall, these patterns suggest that divergent natural selection has played a major role in the ecotypic diversification of *S. lautus*. Further work in confirming edaphic and climatic variables as selective agents requires controlled laboratory experiments.

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

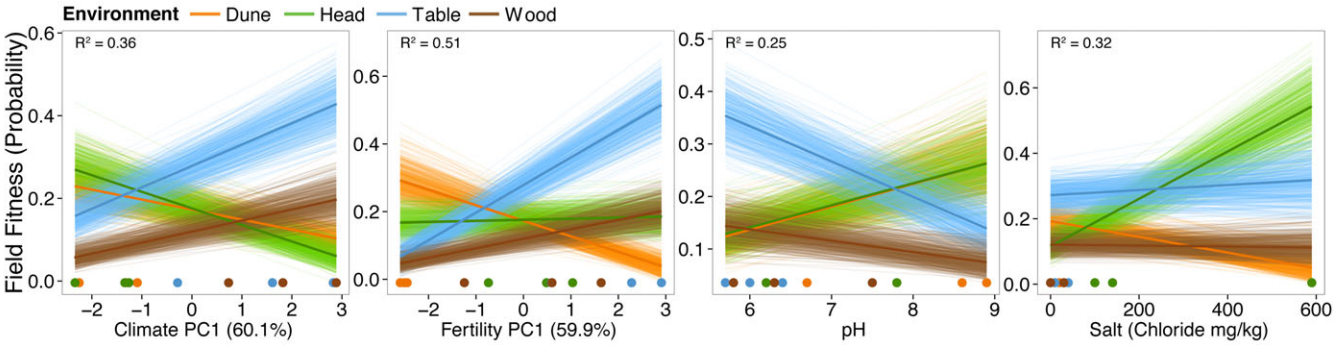


Figure 6. Testing the association between environmental variables and field performance using principal components analyses for suites of related variables. Faint lines are separate regressions between each MCMC iteration and environmental distance to provide the uncertainty around the model. The solid lines indicate the point estimate of the slope and intercept in each environment. Dots along the *x*-axis indicate original population measurements with colors indicating the ecotype they are from. Soil and climate variables were correlated with field performance.

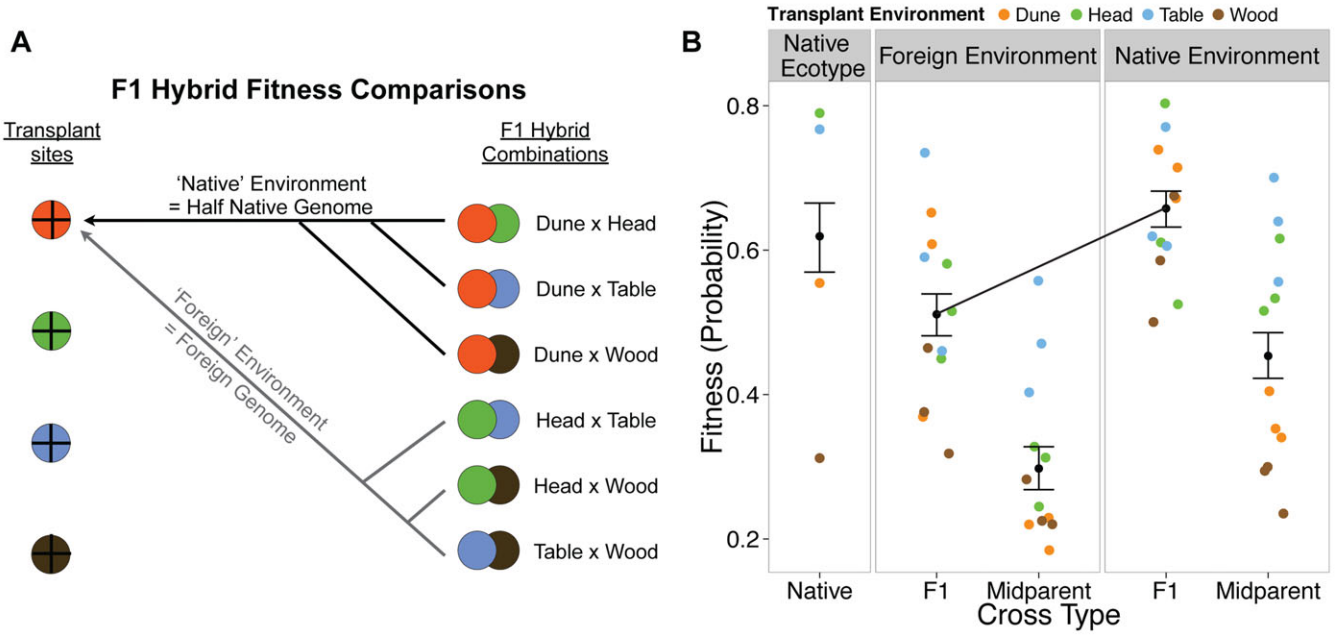


Figure 7. (A) F1 hybrid comparisons when transplanted into a "native" environment (half their genome from a native parental) versus "foreign" environment in which they have no native parent. (B) Performance of F1 hybrids across environments. F1 hybrids with a native parent performed better than hybrids with only foreign parents (diagonal line). F1 hybrids consistently outperformed their midparents and often performed better than or equal to the native ecotype, providing strong evidence for heterosis. Colored dots illustrate the different F1 crosses, with color indicating the environment in which they were transplanted. Confidence intervals are 95% HPD intervals.

CONSEQUENCES FOR EXTRINSIC REPRODUCTIVE ISOLATION

Adaptive divergence is expected to create trade-offs in performance between environments, reducing gene flow and creating extrinsic reproductive isolation (Nosil et al. 2005; Hereford 2009). Immigrant inviability has been uncovered in a variety of systems (reviewed in Lowry et al. 2008a; Baack et al. 2015), but is rarely calculated between more than two populations. Our data suggest that if migration occurs across the heterogeneous landscape, selection against migrants would likely prevent gene flow. We also showed that F1 hybrid crosses carrying 50% of local genes performed better than those F1 hybrids carrying 0% local genes. This result indicates that, concordant with transplant experiments conducted only in the coastal environments (Richards and Ortiz-Barrientos 2016; Richards et al. 2016), extrinsic postzygotic isolation is evolving despite the high levels of heterosis in F1 hybrids. Higher fitness in F1 hybrids with a native parent also suggests that the genes controlling heterosis and those conferring

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

adaptation to the local conditions may be similar. One possibility is that the same dominant alleles may have contributed to the evolution of extrinsic reproductive isolation and heterosis in these ecotypes (Lynch 1991), and that gene flow is likely absent in the system. This would be consistent with the idea that selection tends to be more effective in increasing the frequency of dominant alleles (Whitlock 2003). Whether these alleles came from standing genetic variation or new mutation remains to be explored (Orr and Betancourt 2001).

CONSEQUENCES FOR INTRINSIC REPRODUCTIVE ISOLATION

Whether ecotypes transition to biological species remains contentious (Lowry 2012). Therefore, understanding how genetic incompatibilities arise as a by-product of adaptive divergence remains a core goal in evolutionary biology (Schluter and Conte 2009; Scarpino et al. 2013). Our data are consistent with previous research where initial hybridization leads to hybrid vigor but reduced fitness in the following F2 generation. The initial round of recombination in F2 hybrids disrupts native genetic interactions, reducing intrinsic fitness (Edmands 1999; Stelkens et al. 2015). The recovery of fitness in the F3 hybrids indicates that defunct allelic combinations likely have been removed from the F2 generation and suggests that few but strong negative epistatic interactions are responsible for intrinsic reproductive isolation in this system (Edmands 1999; Fishman and Willis 2001; Stelkens et al. 2015). Given our experimental populations are outbred, a small fraction of F2 genetic incompatibilities would have segregated in the F1 individuals planted in the field, perhaps contributing to the differences in fitness we observed between F1 hybrids with one native versus only foreign parents. Therefore, we conjecture that some of the incompatibilities responsible for reductions in fitness revealed in F2 hybrids arose from the breakdown of functional interactions among genes adapted to specific environments. This would suggest that natural selection not only creates local adaptation but also leads to the evolution of intrinsic reproductive isolation during adaptive diversification.

Conclusion

Our data suggest that where organisms colonize a heterogeneous landscape they respond to specific aspects of the environment, creating ecological divergence. However, recombination between contrasting ecotypes of *S. lautus* reduced intrinsic fitness, suggesting they are accumulating genetic incompatibilities. Therefore, the presence of strong extrinsic reproductive isolation, weak intrinsic reproductive isolation, and geographic separation between ecotypes suggest that adaptive diversification across a heterogeneous landscape is leading to the formation of distinct, irreversible species.

ACKNOWLEDGMENTS

We would like to thank Ruth and Andrew Shaw, Mal Smith, and Ballina Shire Council for generously providing us access to their property for the reciprocal transplant. We are grateful to Maria Servedio, Matthew Streisfeld, and two anonymous reviewers for reviewing this manuscript and providing invaluable feedback that improved the quality of our work. John Dwyer provided comments on previous versions of the manuscript. Libby Liggins, A. Smith, and M. Pawley helped with suggestions during analysis. We are very grateful to everyone who volunteered with fieldwork including J. Walter, L. Ambrose, B. Ayalon, S. Carrol, M. Gallo, F. Roda, A. Maynard, C. Palmer, and H. North. Special thanks to S. Edgley for her repeated dedication to fieldwork. James Donohoe for helping conduct crosses and produces seeds in the glasshouse. Also thanks to K. Marr for extracting the climate variables from GIS layers. This project was funded by Australian Research Council grant DP0986172. Seeds were collected in NSW national parks under permit number S12084.

G.W. and D.O. designed the research; G.W. conducted glasshouse work and prepared all field material; G.W. and T.R. collected soil samples; G.W., M.W., M.J., and T.R. conducted field transplant; G.W. and J.D.A. analyzed data with input from D.O.; G.W. and D.O. wrote the manuscript with all other authors contributing to revisions. All authors declare no conflict of interest.

DATA ARCHIVING

The doi for our data is 10.5061/dryad.d9f36.

LITERATURE CITED

Abbott, R. J., and H. P. Comes. 2007. Blowin' in the wind—the transition from ecotype to species. New Phytol. 175:197–200.

Ågren, J., and D. W. Schemske. 2012. Reciprocal transplants demonstrate strong adaptive differentiation of the model organism *Arabidopsis thaliana* in its native range. New Phytol. 194:1112–1122.

Ali, S. I. 1964. *Senecio lautus* complex in Australia. 2. Cultural studies of populations. Aust. J. Bot. 12:292–316.

Anderson, M. J. 2006. Distance-based tests for homogeneity of multivariate dispersions. Biometrics 62:245–253.

Angert, A. L., H. D. Bradshaw, and D. W. Schemske. 2008. Using experimental evolution to investigate geographic range limits in monkeyflowers. Evolution 62:2660–2675.

Angert, A. L., and D. W. Schemske. 2005. The evolution of species' distributions: reciprocal transplants across the elevation ranges of *Mimulus cardinalis* and *M. lewisii*. Evolution 59:1671–1684.

Antonovics, J., and A. D. Bradshaw. 1970. Evolution in closely adjacent plant populations .VIII. Clinal patterns at a mine boundary. Heredity 25:349–362.

Baack, E., M. C. Melo, L. H. Rieseberg, and D. Ortiz-Barrientos. 2015. The origins of reproductive isolation in plants. New Phytol. 207:968–984.

Baythavong, B. S. 2011. Linking the spatial scale of environmental variation and the evolution of phenotypic plasticity: selection favors adaptive plasticity in fine-grained environments. Am. Nat. 178:75–87.

Baythavong, B. S., and M. L. Stanton. 2010. Characterizing selection on phenotypic plasticity in response to natural environmental heterogeneity. Evolution 64:2904–2920.

Bischoff, A., and H. Muller-Scharer. 2010. Testing population differentiation in plant species—how important are environmental maternal effects. Oikos 119:445–454.

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bridle, J. R., and T. H. Vines. 2007. Limits to evolution at range margins: when and why does adaptation fail? Trends Ecol. Evol. 22:140–147.

Coyne, J. A., and H. A. Orr. 2004. Speciation. Sinauer Associates, Sunderland, MA.

Edmands, S. 1999. Heterosis and outbreeding depression in interpopulation crosses spanning a wide range of divergence. Evolution 53:1757–1768.

Egan, S. P., and D. J. Funk. 2009. Ecologically dependent postmating isolation between sympatric host forms of *Neochlamisus bebbianae* leaf beetles. Proc. Natl. Acad. Sci. USA 106:19426–19431.

Egan, S. P., and J. R. Ott. 2007. Host plant quality and local adaptation determine the distribution of a gall-forming herbivore. Ecology 88:2868–2879.

Endler, J. A. 1977. Geographic variation, speciation, and clines. Princeton Univ. Press, Princeton, New Jersey.

Fenster, C. B., and L. F. Galloway. 2000. Population differentiation in an annual legume: genetic architecture. Evolution 54:1157–1172.

Fishman, L., and J. H. Willis. 2001. Evidence for Dobzhansky–Muller incompatibilites contributing to the sterility of hybrids between *Mimulus guttatus* and *Mimulus nasutus*. Evolution 55:1932–1942.

Fry, J. D. 2003. Detecting ecological trade-offs using selection experiments. Ecology 84:1672–1678.

Gandon, S., and S. L. Nuismer. 2009. Interactions between genetic drift, gene flow, and selection mosaics drive parasite local adaptation. Am. Nat. 173:212–224.

Gelman, A. 2006. Prior distributions for variance parameters in hierarchical models. Bayesian Anal. 1:515–533.

Hadfield, J. D. 2010. MCMC methods for multi-response generalized linear mixed models: the MCMCglmm R package. J. Stat. Softw. 33:1–22.

Hall, M. C., D. B. Lowry, and J. H. Willis. 2010. Is local adaptation in *Mimulus guttatus* caused by trade-offs at individual loci? Mol. Ecol. 19:2739–2753.

Hatfield, T., and D. Schluter. 1999. Ecological speciation in sticklebacks: environment-dependent hybrid fitness. Evolution 53:866–873.

Hereford, J. 2009. A quantitative survey of local adaptation and fitness trade-offs. Am. Nat. 173:579–588.

Hereford, J., and A. A. Winn. 2008. Limits to local adaptation in six populations of the annual plant *Diodia* teres. New Phytol. 178:888–896.

Johansen-Morris, A. D., and R. G. Latta. 2006. Fitness consequences of hybridization between ecotypes of *Avena barbata*: hybrid breakdown, hybrid vigor, and transgressive segregation. Evolution 60:1585–1595.

Joshi, J., B. Schmid, M. C. Caldeira, P. G. Dimitrakopoulos, J. Good, R. Harris, A. Hector, K. Huss-Danell, A. Jumpponen, A. Minns, et al. 2001. Local adaptation enhances performance of common plant species. Ecol. Lett. 4:536–544.

Kawecki, T. J., and D. Ebert. 2004. Conceptual issues in local adaptation. Ecol. Lett. 7:1225–1241.

Lenormand, T. 2002. Gene flow and the limits to natural selection. Trends Ecol. Evol. 17:183–189.

Lowry, D. B. 2012. Ecotypes and the controversy over stages in the formation of new species. Biol. J. Linn. Soc. 106:241–257.

Lowry, D. B., K. D. Behrman, P. Grabowski, G. P. Morris, J. R. Kiniry, and T. E. Juenger. 2014. Adaptations between ecotypes and along environmental gradients in *Panicum virgatum*. Am. Nat. 183:682–692.

Lowry, D. B., M. C. Hall, D. E. Salt, and J. H. Willis. 2009. Genetic and physiological basis of adaptive salt tolerance divergence between coastal and inland *Mimulus guttatus*. New Phytol. 183:776–788.

Lowry, D. B., J. L. Modliszewski, K. M. Wright, C. A. Wu, and J. H. Willis. 2008a. The strength and genetic basis of reproductive isolating barriers in flowering plants. Philos. T Roy. Soc. B 363:3009–3021.

Lowry, D. B., R. C. Rockwood, and J. H. Willis. 2008b. Ecological reproductive isolation of coast and inland races of *Mimulus guttatus*. Evolution 62:2196–2214.

Lynch, M. 1991. The genetic interpretation of inbreeding depression and outbreeding depression. Evolution 45:622–629.

MacColl, A. D. C. 2011. The ecological causes of evolution. Trends Ecol. Evol. 26:514–522.

Macel, M., C. S. Lawson, S. R. Mortimer, M. Smilauerova, A. Bischoff, L. Cremieux, J. Dolezal, A. R. Edwards, V. Lanta, T. M. Bezemer, et al. 2007. Climate vs. soil factors in local adaptation of two common plant species. Ecology 88:424–433.

Melo, M. C., A. Grealy, B. Brittain, G. M. Walter, and D. Ortiz-Barrientos. 2014. Strong extrinsic reproductive isolation between parapatric populations of an Australian groundsel. New Phytol. 203:323–334.

Melo-Hurtado, M. C. 2014. Ecological speciation in *Senecio lautus*. PhD Thesis. School of Biological Sciences, University of Queensland, St. Lucia, Australia.

Nosil, P. 2012. Ecological speciation. Oxford Univ. Press, Oxford.

Nosil, P., T. H. Vines, and D. J. Funk. 2005. Reproductive isolation caused by natural selection against immigrants from divergent habitats. Evolution 59:705–719.

Oksanen, J., F. G. Blanchet, R. Kindt, P. Legendre, P. R. Minchin, R. B. O'Hara, G. L. Simpson, P. Solymos, M. H. H. Stevens, and H. Wagner. 2016. vegan: community ecology package. R package version 2.3-3, Available at https://CRAN.R-project.org/package=vegan.

Orr, H. A., and A. J. Betancourt. 2001. Haldane's sieve and adaptation from the standing genetic variation. Genetics 157:875–884.

Orr, M. R., and T. B. Smith. 1998. Ecology and speciation. Trends Ecol. Evol. 13:502–506.

Peterson, M. L., K. M. Kay, and A. L. Angert. 2016. The scale of local adaptation in *Mimulus guttatus*: comparing life history races, ecotypes, and populations. New Phytol. 211:345–356.

Radford, I. J., R. D. Cousens, and P. W. Michael. 2004. Morphological and genetic variation in the *Senecio pinnatifolius* complex: are variants worthy of taxonomic recognition? Aust. Syst. Bot. 17:29–48.

Rhode, J. M., and M. B. Cruzan. 2005. Contributions of heterosis and epistasis to hybrid fitness. Am. Nat. 166:E124–E139.

Richards, T. J., and D. Ortiz-Barrientos. 2016. Immigrant inviability produces a strong barrier to gene flow between parapatric ecotypes of *Senecio lautus*. Evolution 70:1239–1248.

Richards, T. J., G. M. Walter, K. McGuigan, and D. Ortiz-Barrientos. 2016. Divergent natural selection drives the evolution of reproductive isolation in an Australian wildflower. Evolution. Accepted Author Manuscript. doi: 10.1111/evo.12994.

Roda, F., L. Ambrose, G. M. Walter, H. L. L. Liu, A. Schaul, A. Lowe, P. B. Pelser, P. Prentis, L. H. Rieseberg, and D. Ortiz-Barrientos. 2013a. Genomic evidence for the parallel evolution of coastal forms in the *Senecio lautus* complex. Mol. Ecol. 22:2941–2952.

Roda, F., H. L. Liu, M. J. Wilkinson, G. M. Walter, M. E. James, D. M. Bernal, M. C. Melo, A. Lowe, L. H. Rieseberg, P. Prentis, and D. Ortiz-Barrientos. 2013b. Convergence and divergence during the adaptation to similar environments by an Australian groundsel. Evolution 67:2515–2529.

Rundle, H. D. 2002. A test of ecologically dependent postmating isolation between sympatric sticklebacks. Evolution 56:322–329.

Rundle, H. D., and M. C. Whitlock. 2001. A genetic interpretation of ecologically dependent isolation. Evolution 55:198–201.

Sambatti, J. B. M., and K. J. Rice. 2007. Functional ecology of ecotypic differentiation in the Californian serpentine sunflower (*Helianthus exilis*). New Phytol. 175:107–119.

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Scarpino, S. V., P. J. Hunt, F. J. Garcia-De-Leon, T. E. Juenger, M. Schartl, and M. Kirkpatrick. 2013. Evolution of a genetic incompatibility in the genus *Xiphophorus*. Mol. Biol. Evol. 30:2302–2310.

Schluter, D. 2001. Ecology and the origin of species. Trends Ecol. Evol. 16:372–380.

Schluter, D., and G. L. Conte. 2009. Genetics and ecological speciation. Proc. Natl. Acad. Sci. USA 106:9955–9962.

Seehausen, O., R. K. Butlin, I. Keller, C. E. Wagner, J. W. Boughman, P. A. Hohenlohe, C. L. Peichel, G. P. Saetre, C. Bank, A. Brannstrom, et al. 2014. Genomics and the origin of species. Nat. Rev. Genet. 15:176–192.

Selby, J. 2014. The genetic basis of local adaptation to serpentine soils in *Mimulus guttatus*. PhD Thesis. Department of Biology, Duke University, Durham, NC.

Simpson, G. G. 1953. The major features of evolution. Columbia Univ. Press, New York.

Steinsland, I., C. T. Larsen, A. Roulin, and H. Jensen. 2014. Quantitative genetic modeling and inference in the presence of nonignorable missing data. Evolution 68:1735–1747.

Stelkens, R. B., C. Schmid, and O. Seehausen. 2015. Hybrid breakdown in cichlid fish. Plos One 10:e0127207.

Therneau, T. 2015. Mixed effects cox models. R package version 2.2.3. Available at: http://CRAN.R-project.org/package=coxme.

Thompson, I. R. 2005. Taxonomic studies of Australian *Senecio* (Asteraceae): 5. The *S. pinnatifolius/S. lautus* complex. Muelleria 21:23–76.

Vines, T. H., and D. Schluter. 2006. Strong assortative mating between allopatric sticklebacks as a by-product of adaptation to different environments. P Roy Soc. Lond. B Bio. 273:911–916.

Whitlock, M. C. 2003. Fixation probability and time in subdivided populations. Genetics 164:767–779.

Associate Editor: M. Streisfeld
Handling Editor: M. Servedio

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. Population locations. Coloured rows indicate the transplant sites.
Table S2. Numbers of individuals and full-sibling families used to create late generation hybrids.
Figure S3. Photos of seedlings and plots in the transplant environments.
Supplementary material S4. R code for the bivariate model.
Figure S5. Correlations between the first principal components conducted on the major soil environmental variables.
Figure S6. Regressions for the second climate and soil fertility principal components and fitness.
Figure S7. A summary of the data taken from the weather station placed at each transplant location.
Figure S8. Test of whether geographic distance from each transplant site explained fitness. Low R2 values show a poor fit to the data, suggesting that geographic distance was not associated with fitness.
Figure S9. Significance test for the regression slopes.

15585646, 2016, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/evo.13009 by The University Of Queensland, Wiley Online Library on [10/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License